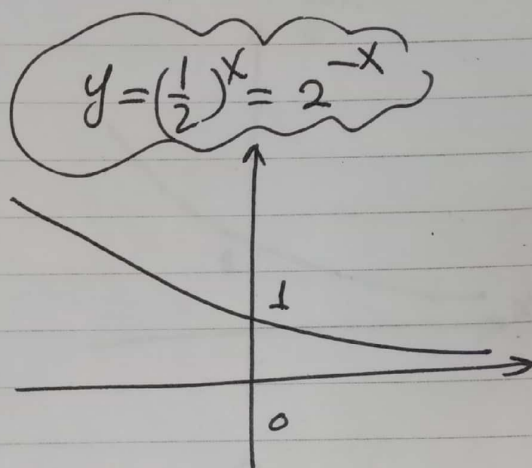
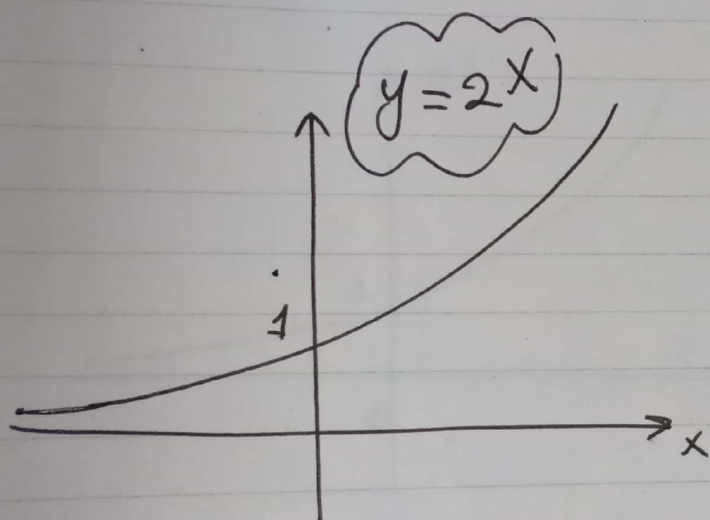


Calculus I - Fall 2019

Lecture #2

Exponential and Logarithmic Functions

Exponential Function: $y = a^x$ ($a > 0$)
 a : constant



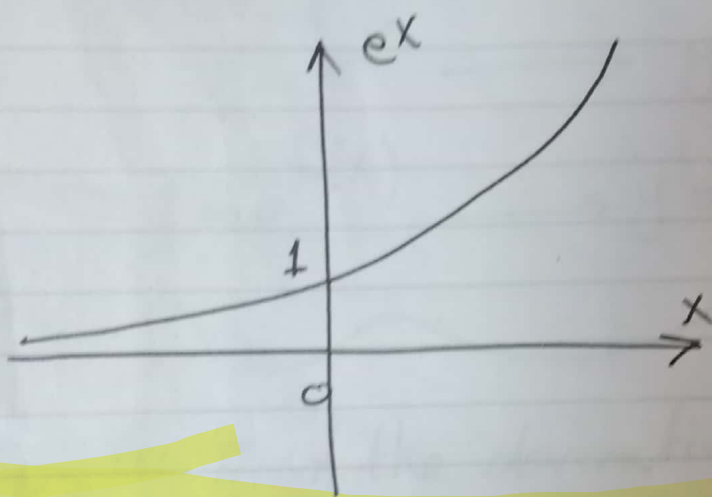
Derivative: $y(x) = a^x$

$$y'(x) = \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \frac{0}{0}$$
$$= \lim_{h \rightarrow 0} a^x \frac{a^h - 1}{a} = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

$\downarrow$
this limit goes to "1"
at $a = 2.7182818 \dots$

Let $e = 2.7182818 \dots$
then we have $y = e^x \rightarrow y' = e^x$

$$f(x) = e^x \Rightarrow f'(x) = e^x$$

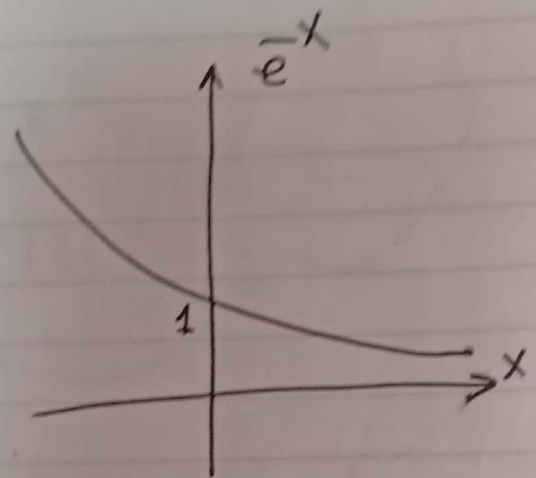


Domain: $\mathbb{R}$

Range: $]0, \infty[$

$$\lim_{x \rightarrow \infty} e^x = \infty$$

$$\lim_{x \rightarrow -\infty} e^x = 0$$



Domain: $\mathbb{R}$

Range: $]0, \infty[$

$$\lim_{x \rightarrow \infty} e^{-x} = 0$$

$$\lim_{x \rightarrow -\infty} e^{-x} = \infty$$

Remark: The exponential function doesn't return a non +ve value.

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Example: Find the domain of

a) $y = e^{2x^2 - 5x + 1}$

b) $y = e^{\frac{1}{x+1}}$

c) $y = e^{\sqrt{x}}$

Solution:

a) $D: \mathbb{R}$

b) $D: \mathbb{R} - \{-1\}$

c) $D: [0, \infty[$

Rule:

$$y = e^x \longrightarrow y' = e^x$$

$$y = e^{f(x)} \longrightarrow y' = f'(x) e^{f(x)}$$

Example: Find the derivative of

a) $y = e^{5x^2 - 10x + 1}$

b) $y = e^{\sqrt{x}}$

c) $y = e^{\frac{x}{x-1}}$

d) $y = (2e^{-x} - x)^4$

e) $y = e^x + x^e$

f) $y = e^{x^2+1} - (x^2+1)^e$

Solution:

a) $y' = (10x - 10) e^{5x^2 - 10x + 1}$

b) $y' = \frac{1}{2\sqrt{x}} e^{\sqrt{x}}$

c) $y' = \frac{(x-1) - x}{(x-1)^2} e^{\frac{x}{x-1}}$

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$$d) y' = 4(2e^{-x} - x)^3 (-2e^{-x} - 1)$$

$$e) y' = e^x + e x^{e-1}$$

$$f) y' = 2x e^{x^2+1} - e(x^2+1)^{e-1} (2x)$$

Logarithms:

Recall that $\log_{10} 1000 = 3$ $\log_2 16 = 4$ $\log_3 \frac{1}{9} = -2$

$$\log_{217} 217 = 1$$

$$\log_{1039} 1 = 0$$

$$\log_{10}(-10) \text{ "Undefined"}$$

If the base of the logarithm is the constant "e", we call it "ln"

$$\text{So, } \ln \sqrt{e} = \frac{1}{2} \quad \ln \frac{1}{e^2} = -2$$

$$\ln \frac{1}{\sqrt[3]{e}} = -\frac{1}{3} \quad \ln e = 1 \quad \ln 1 = 0$$

$\ln(-e)$ is undefined.

$$\ln e = 1$$

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Notes:

$$\begin{aligned}y &= \ln x \rightarrow x = e^y \\ \ln u &= v \rightarrow u = e^v \\ \theta &= e^\phi \rightarrow \phi = \ln \theta \\ e^s &= t \rightarrow s = \ln t\end{aligned}$$

Example: solve for x

a) $12e^x + 3 = 7$

b) $e^{2x-3} = 4$

c) $5 - 2\ln x = 1$

d) $\ln(1-2x) = 2$

solution:

a) $12e^x + 3 = 7 \rightarrow e^x = \frac{1}{3} \rightarrow x = \ln \frac{1}{3} = -1.09$

b) $e^{2x-3} = 4 \rightarrow 2x-3 = \ln 4 \rightarrow x = \frac{3+\ln 4}{2} = 2.19$

c) $5 - 2\ln x = 1 \rightarrow \ln x = 2 \rightarrow x = e^2 = 7.389$

d) $\ln(1-2x) = 2 \rightarrow 1-2x = e^2 \rightarrow x = \frac{1-e^2}{2} = -3.195$

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Rule

$$y = a^x \longrightarrow y' = a^x \ln a$$

$$y = a^{f(x)} \longrightarrow y' = a^{f(x)} \ln a \cdot f'(x)$$

Examples Find the derivative of

a) $y = 4^x$

b) $y = 10^x + x^{10}$

c) $y = 8^{x^2}$

d) $y = 6^{\sqrt{x^2+1}}$

e) $y = 3^{x^3-2x+1} + (x^3-2x+1)^3$

Solution:

a) $y' = 4^x \ln 4$

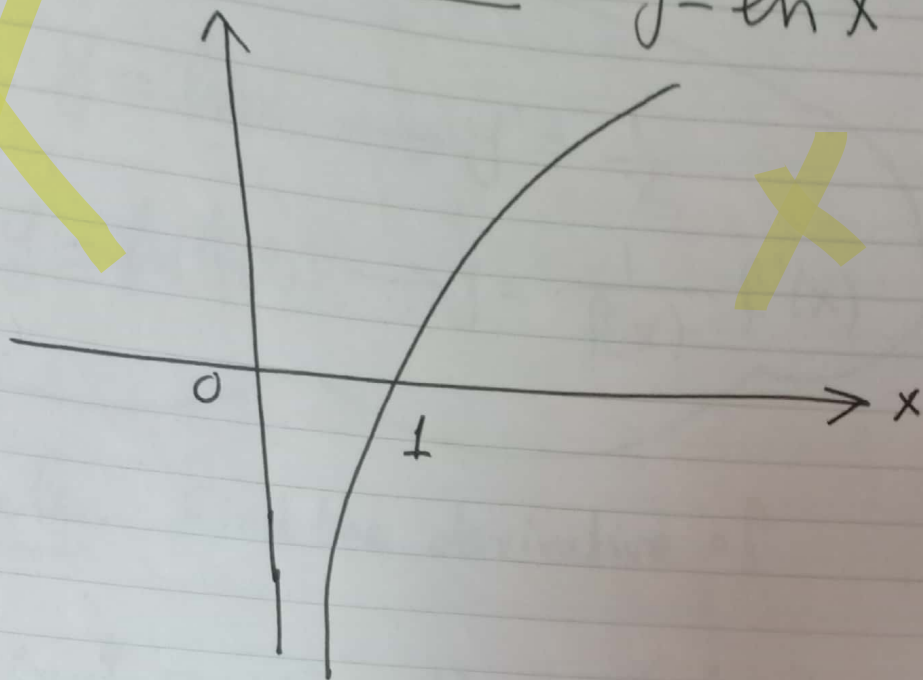
b) $y' = 10^x \ln 10 + 10x^9$

c) $y' = 8^{x^2} \ln 8 (2x)$

d) $y' = 6^{\sqrt{x^2+1}} \ln 6 \left(\frac{2x}{2\sqrt{x^2+1}} \right)$

e) $y' = 3^{x^3-2x+1} \ln 3 (3x^2-2) + 3(x^3-2x+1)^2 (3x^2-2)$

The Logarithmic Function: $y = \ln x$



Domain: $]0, \infty[$

Range: $\mathbb{R}$

$$\lim_{x \rightarrow \infty} \ln x = \infty, \quad \lim_{x \rightarrow 0^+} \ln x = -\infty$$

Notice that: $\ln 1 = 0$

$$\ln 0 \neq 1 \Rightarrow \ln 0 = -\infty$$

Example: Find the domain of $y = \ln(2x-1)$

Solution: $2x-1 > 0 \rightarrow x > \frac{1}{2}$

$$D:]\frac{1}{2}, \infty[$$

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Rule

$$y = \ln x \rightarrow y' = \frac{1}{x}$$

$$y = \ln(f(x)) \rightarrow y' = \frac{1}{f(x)} f'(x)$$

Examples Find the derivative of

a) $y = \ln(3x^4 - 5x + 1)$ b) $y = \ln\left(\frac{x+1}{x-1}\right)$

c) $y = \sqrt{\ln(10e^x - 4^x + 1)}$

Solutions

a) $y' = \frac{12x^3 - 5}{3x^4 - 5x + 1}$

b) $y' = \frac{x-1}{x+1} \cdot \frac{(x-1) - (x+1)}{(x-1)^2} = \frac{-2}{(x+1)(x-1)}$
 $= \frac{-2}{x^2 - 1}$

c) $y = (\ln(10e^x - 4^x + 1))^{1/2}$

$$y' = \frac{1}{2\sqrt{\ln(10e^x - 4^x + 1)}} \cdot \frac{1}{10e^x - 4^x + 1} (10e^x - 4^x \ln 4)$$

Properties of the logarithm:

$$\ln(AB) = \ln A + \ln B$$

$$\ln\left(\frac{A}{B}\right) = \ln A - \ln B$$

$$\ln(A^B) = B \ln A$$

$$\ln(e^B) = B, \quad e^{\ln B} = B$$

Example: Find y' if $y = \ln\left(\frac{x(x^2+1)^4}{\sqrt{3x-1}}\right)$

Solution:

$$y = \ln(x) + \ln(x^2+1)^4 - \ln\sqrt{3x-1}$$

$$y = \ln(x) + 4 \ln(x^2+1) - \frac{1}{2} \ln(3x-1)$$

$$\Rightarrow y' = \frac{1}{x} + 4 \frac{2x}{x^2+1} - \frac{1}{2} \frac{3}{3x-1}$$

$$= \frac{1}{x} + \frac{8x}{x^2+1} - \frac{3}{2(3x-1)}$$

Example: $y = \sqrt{\frac{(x^6+1)^5 e^{\frac{1}{x}}}{3x^2 - x + 5^x + 1}}$

Find y'

Solution:

$$y = \left(\frac{(x^6+1)^5 e^{\frac{1}{x}}}{3x^2 - x + 5^x + 1} \right)^{\frac{1}{2}}$$

$$\ln y = \frac{1}{2} \left[\ln(x^6+1)^5 + \ln e^{\frac{1}{x}} - \ln(3x^2 - x + 5^x + 1) \right]$$

$$\ln y = \frac{1}{2} \left[5 \ln(x^6+1) + \frac{1}{x} \ln e - \ln(3x^2 - x + 5^x + 1) \right]$$

Differentiate w.r.t x

$$\frac{1}{y} y' = \frac{1}{2} \left[5 \frac{6x^5}{x^6+1} - \frac{1}{x^2} - \frac{6x-1+5^x \ln 5}{3x^2 - x + 5^x + 1} \right]$$

$$\Rightarrow y' = \frac{y}{2} \left[\frac{30x^5}{x^6+1} - \frac{1}{x^2} - \frac{6x-1+5^x \ln 5}{3x^2 - x + 5^x + 1} \right]$$
